UNIT 4: PHOTOSYNTHESIS AND CELLULAR ENERGETICS
Course: BIO201 Plant Physiology | Department of Biological Sciences

1. Introduction to Photosynthesis
Photosynthesis is defined as the metabolic process by which photoautotrophic organisms convert light energy into chemical e
The overall balanced equation is: 6CO2 + 6H2O + light -> C6H12O6 + 6O2.
This critical biochemical transformation takes place within specialized double-membraned plant organelles known as chloropla

2. Light-Dependent Reactions
Light-Dependent Reactions occur across the thylakoid membrane system.
Photons excite chlorophyll pigments in Photosystem II (PSII, P680) and Photosystem I (PSI, P700).
Photolysis of water generates protons, electrons, and molecular oxygen (O2).
The established proton motive force across the thylakoid lumen drives ATP synthase phosphorylation, yielding ATP and reduc

3. The Calvin-Benson Cycle (Light-Independent Phase)
The Calvin Cycle operates within the aqueous stroma of the chloroplast.
Phase 1 (Carbon Fixation): Atmospheric carbon dioxide is catalyzed by the key enzyme RuBisCO.
RuBisCO fixes CO_2 with ribulose-1,5-bisphosphate to form 3-phosphoglycerate (3-PGA).
Phase 2 (Reduction): ATP and NADPH reduce 3-PGA into glyceraldehyde-3-phosphate (G3P).
Phase 3 (Regeneration): G3P molecules are recycled to regenerate RuBP, completing the cyclic pathway.

4. Comparative Evaluation: Light Reactions vs Calvin Cycle
Light Reactions require direct photon excitation, take place in thylakoids, and generate ATP, NADPH, and O_2.
The Calvin Cycle takes place in the stroma, consumes ATP and NADPH, and synthesizes carbohydrates without direct light.